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La Palma Earthquakes Mechanisms

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Abstract

In September 2021, a significant jump in seismic activity on the island of La Palma (Canary Islands, Spain) signaled the start of a volcanic crisis that still continues at the time of writing. Earthquake data is continually collected and published by the Instituto Geográfico Nacional (IGN). We aimed at describing alternative methods to describe the probability of a new eruption considering historical data.

Plain Language Summary

Earthquake data for the island of La Palma from the September 2021 eruption is found in the database of the Instituto Geográfico Nacional (IGN). Current methods of statistical probability have limited advantages in the prediction of new eruptions.

1 Introduction

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Figure 1: Timeline of recent earthquakes on La Palma

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Based on data up to and including 2021, eruptions on La Palma happen every 79.8 years on average.

Studies of the magma systems feeding the volcano, such as Marrero et al. (2019), have proposed that there are two main magma reservoirs feeding the Cumbre Vieja volcano; one in the mantle (30-40km depth) which charges and in turn feeds a shallower crustal reservoir (10-20km depth).

Eight eruptions have been recorded since the late 1400s (Figure 1).

Data and methods are discussed in Section 2.

Let x denote the number of eruptions in a year. Then, x can be modeled by a Poisson distribution

$$p(x) = \frac{e^{-\lambda} \lambda^x}{x!} \quad (1)$$

where λ is the rate of eruptions per year. Using Equation 1, the probability of an eruption in the next t years can be calculated.

Table 1: Recent historic eruptions on La Palma

Name	Year
Current	2021
Teneguía	1971
Nambroque	1949
El Charco	1712
Volcán San Antonio	1677
Volcán San Martín	1646
Tajuya near El Paso	1585
Montaña Quemada	1492

31 Table 1 summarises the eruptions recorded since the colonization of the islands by
 32 Europeans in the late 1400s.

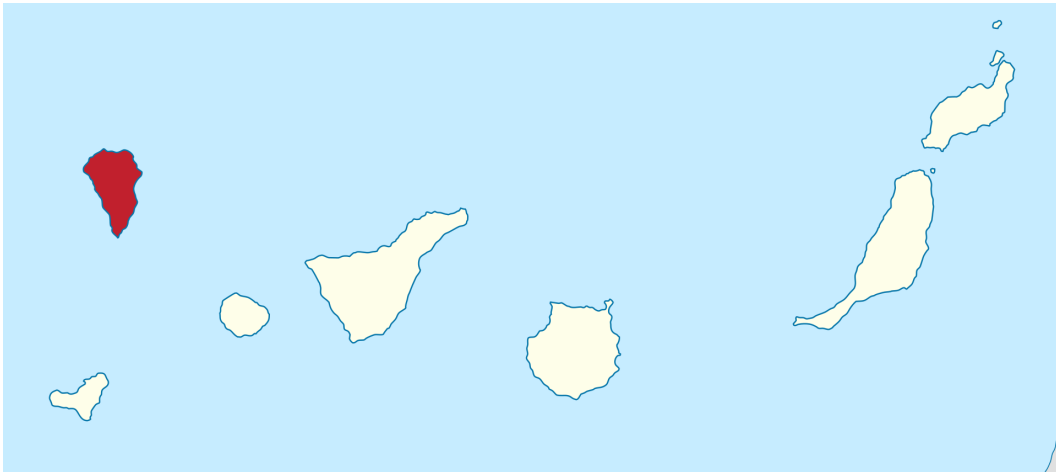


Figure 2: Map of La Palma

33 La Palma is one of the west most islands in the Volcanic Archipelago of the Canary
 34 Islands (Figure 2).

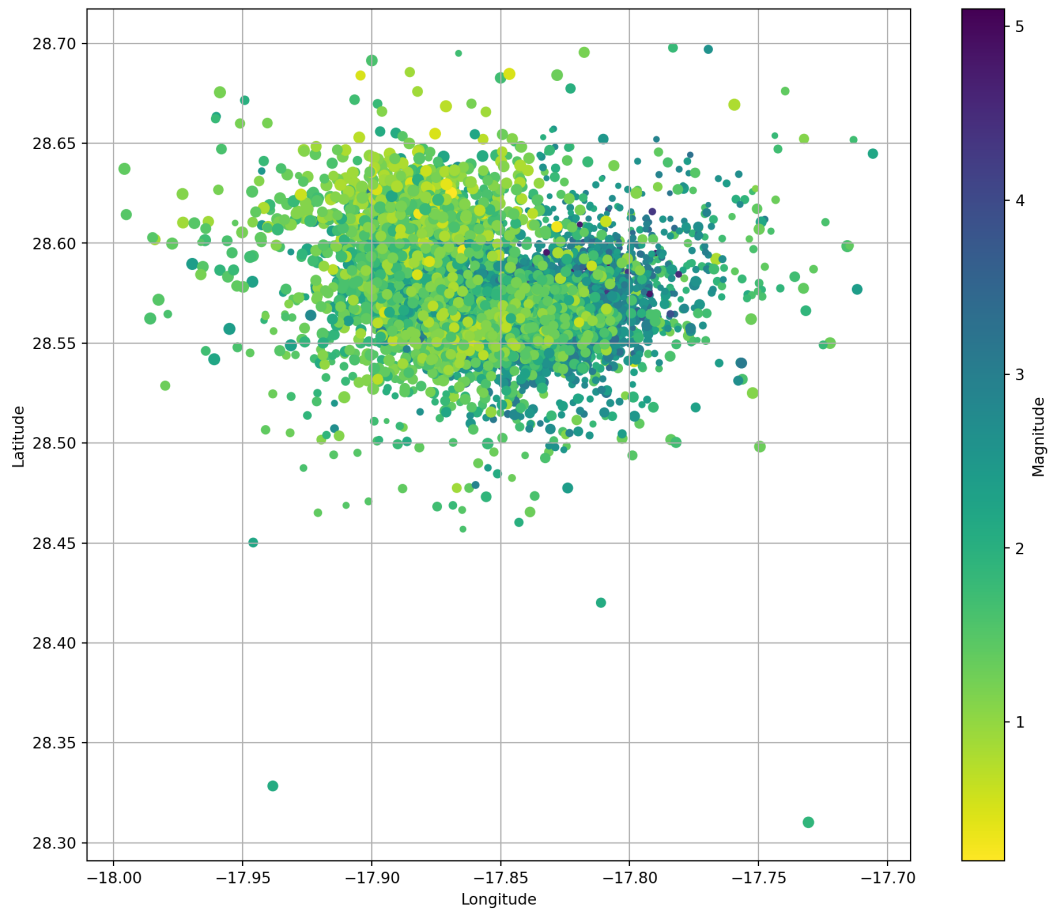


Figure 3: Earthquakes on La Palma since 2017.

Source: [Article Notebook](#)

Figure 3 shows the location of recent Earthquakes on La Palma.

2 Data & Methods

3 Conclusion

References

Source: [Article Notebook](#)

Marrero, J., García, A., Berrocoso, M., Llinares, Á., Rodríguez-Losada, A., & Ortiz, R. (2019). Strategies for the development of volcanic hazard maps in monogenetic volcanic fields: The example of La Palma (Canary Islands). *Journal of Applied Volcanology*, 8. <https://doi.org/10.1186/s13617-019-0085-5>